6. Strings

1. Different ways creating a string

Answer:

```
public class StringCreation {
    public static void main(String[] args) {

        // String literal
        String str1 = "Hello";

        // Using new keyword
        String str2 = new String("World");

        // Using character array
        char[] chars = {'J', 'A', 'V', 'A'};
        String str3 = new String(chars);

        // Print strings
        System.out.println(str1);
        System.out.println(str2);
        System.out.println(str3);
    }
}
```

2. Concatenating two strings using + operator

Answer:

```
public class StringConcatenation {
    public static void main(String[] args) {

        // Declare strings
        String firstName = "Suman";
        String lastName = "Kumar";

        // Concatenate strings
        String fullName = firstName + " " + lastName;

        // Print result
        System.out.println("Full Name: " + fullName);
    }
}
```

3. Finding the length of the string

Answer:

```
public class StringLength {
    public static void main(String[] args) {

        // Create string
        String text = "Java Programming";

        // Find length
        int length = text.length();

        // Print length
        System.out.println("Length of string: " + length);
```

```
    }
}
```

4. Extract a string using Substring

Answer:

```
public class StringSubstring {
    public static void main(String[] args) {

        // Create string
        String text = "Java Programming";

        // Extract substring
        String sub = text.substring(5);

        // Print substring
        System.out.println("Substring: " + sub);
    }
}
```

5. Searching in strings using indexOf()

Answer:

```
public class StringIndexOf {
    public static void main(String[] args) {

        // Create string
        String text = "Java Programming";

        // Find index of character
        int index = text.indexOf("P");

        // Print result
        System.out.println("Index of P: " + index);
    }
}
```

6. Matching a String Against a Regular Expression With matches()

Answer:

```
public class StringMatches {
    public static void main(String[] args) {

        // Create string
        String text = "12345";

        // Check if string contains only numbers
        boolean result = text.matches("[0-9]+");

        // Print result
        System.out.println("Matches: " + result);
    }
}
```

7. Comparing strings using the methods equals(),
Answer:
```
public class StringEquals {
    public static void main(String[] args) {

        // Create strings
        String str1 = "Java";
        String str2 = "Java";

        // Compare strings
        boolean result = str1.equals(str2);

        // Print result
        System.out.println("Equal: " + result);
    }
}
```

8. equalsIgnoreCase(), startsWith(), endsWith() and compareTo()
Answer:
```
public class StringComparisonMethods {
    public static void main(String[] args) {

        // Create strings
        String str1 = "Java";
        String str2 = "java";

        // equalsIgnoreCase()
        System.out.println("equalsIgnoreCase: " +
str1.equalsIgnoreCase(str2));

        // startsWith()
        System.out.println("startsWith Ja: " + str1.startsWith("Ja"));

        // endsWith()
        System.out.println("endsWith va: " + str1.endsWith("va"));

        // compareTo()
        System.out.println("compareTo: " + str1.compareTo(str2));
    }
}
```

9. Trimming strings with trim()
Answer:
```
public class StringTrim {
    public static void main(String[] args) {

        // String with spaces
        String text = "   Java Programming   ";

        // Remove spaces
        String trimmed = text.trim();

        // Print result
```

```
        System.out.println(trimmed);
    }
}
```

10. Replacing characters in strings with replace()

Answer:

```
public class StringReplace {
    public static void main(String[] args) {

        // Create string
        String text = "Java";

        // Replace character
        String newText = text.replace('a', 'o');

        // Print result
        System.out.println("New String: " + newText);
    }
}
```

11. Splitting strings with split()

Answer:

```
public class StringSplit {
    public static void main(String[] args) {

        // Create string
        String text = "Java,Python,C++";

        // Split string
        String[] languages = text.split(",");

        // Print array values
        for (String lang : languages) {
            System.out.println(lang);
        }
    }
}
```

12. Converting Numbers to Strings with valueOf()

Answer:

```
public class NumberToString {
    public static void main(String[] args) {

        // Integer number
        int number = 100;

        // Convert number to string
        String str = String.valueOf(number);

        // Print result
        System.out.println("String value: " + str);
    }
}
```

13. Converting integer objects to Strings

Answer:

```
public class IntegerObjectToString {
    public static void main(String[] args) {

        // Create Integer object
        Integer number = 500;

        // Convert Integer object to String
        String str = number.toString();

        // Print result
        System.out.println("Converted String: " + str);
    }
}
```

14. Converting to uppercase and lowercase

Answer:

```
public class UpperLowerCase {
    public static void main(String[] args) {

        // Create string
        String text = "Java Programming";

        // Convert to uppercase
        String upper = text.toUpperCase();

        // Convert to lowercase
        String lower = text.toLowerCase();

        // Print results
        System.out.println("Uppercase: " + upper);
        System.out.println("Lowercase: " + lower);
    }
}
```